

IFNAR1 Deficiency: A Comprehensive Disease Characteristics Report

Disease: IFNAR1 Deficiency (Autosomal Recessive) **MONDO ID:** MONDO:0030970 · **OMIM:** #619304 (Immunodeficiency 45) · **Gene OMIM:** *IFNAR1* 107450 **Category:** Mendelian inborn error of immunity (IEI)

Summary

IFNAR1 deficiency is an ultra-rare **autosomal recessive inborn error of immunity** caused by biallelic loss-of-function (LOF) variants in *IFNAR1* (chromosome 21q22.11), the gene encoding interferon-alpha/beta receptor subunit 1. IFNAR1 pairs with IFNAR2 to form the shared cell-surface receptor for **all type I interferons (IFN- α , IFN- β , and IFN- ω)**. When IFNAR1 is absent or non-functional, cells cannot respond to any type I IFN, abolishing the JAK1/TYK2-STAT1/STAT2-ISGF3 signaling cascade that drives interferon-stimulated gene (ISG) expression and intrinsic antiviral defense. The functional signature is definitive: patient fibroblasts show no response to IFN- α 2, IFN- ω , or IFN- β [PMID: 35442418](#).

The clinical hallmark is a striking dissociation between baseline health and catastrophic response to specific viral challenges. Affected individuals are typically **otherwise healthy and develop normally** until they encounter a live attenuated vaccine (measles-mumps-rubella, MMR; or yellow fever 17D) or a select wild-type virus (notably SARS-CoV-2), whereupon uncontrolled viral replication produces severe, sometimes fatal disseminated disease — encephalitis, pneumonitis, hepatitis, and multiorgan involvement. This pattern illustrates an **"essential but narrow" role for type I IFN in human antiviral immunity**: unlike mice, humans display considerable redundancy in type I IFN protection under natural conditions, so the phenotype is dominated by attenuated-vaccine viruses and a handful of wild-type pathogens rather than by broad, everyday viral susceptibility.

The disease has a notable population genetics dimension: a **Polynesian founder allele, p.Glu386* (nonsense)**, reaches a minor allele frequency above 1% in Samoa and is distributed across western Polynesia, making regional consideration essential before live vaccination. Diagnosis combines genetic testing (WES/WGS/IEI panels, with copy-number analysis for large

deletions) and functional confirmation of absent type I IFN responses, while carefully excluding the **acquired phenocopy** — neutralizing autoantibodies against type I IFNs, found in 10–15% of critical COVID-19 pneumonia cases. Management is fundamentally preventive: strict avoidance of live attenuated vaccines, supportive/antiviral care during viral episodes, cascade screening of relatives, and hematopoietic stem cell transplantation (HSCT) as a rational curative option in severe disease. There is no approved disease-specific pharmacotherapy, and exogenous type I IFN is useless because the receptor is absent.

1. Disease Information

IFNAR1 deficiency is a Mendelian inborn error of immunity characterized by selective vulnerability to certain viruses and live attenuated viral vaccines, with intact immunity to most other pathogens. It was first described in 2019 in otherwise healthy patients who suffered life-threatening disease after MMR and yellow fever vaccination [PMID: 31270247](#).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0030970
OMIM (phenotype)	#619304 (Immunodeficiency 45)
OMIM (gene)	<i>IFNAR1</i> 107450
Gene (HGNC)	<i>IFNAR1</i> (HGNC:5432)
Cytogenetic locus	21q22.11
Inheritance	Autosomal recessive

Synonyms / alternative names: Immunodeficiency 45 (IMD45); autosomal recessive IFNAR1 deficiency; interferon alpha/beta receptor 1 deficiency; type I interferon receptor 1 deficiency.

Information source: The disease-level knowledge derives primarily from **aggregated case series and functional immunology studies** (Hernandez et al. 2019; Bastard et al. 2022; Abolhassani et al. 2022; Azamor et al. 2026), not from EHR/population-scale data, reflecting the ultra-rare nature of the condition.

"Globally, autosomal recessive IFNAR1 deficiency is a rare inborn error of immunity underlying susceptibility to live attenuated vaccine and wild-type viruses." — PMID: 35442418

2. Etiology

Disease causal factor (genetic): The sole cause is **biallelic loss-of-function variants in IFNAR1**. This is a monogenic, molecularly fully penetrant defect. Cellular responses to all type I IFNs are abolished [PMID: 35442418](#).

Genetic risk factors: The disease is the genotype — homozygous or compound heterozygous LOF *IFNAR1* variants. No additional susceptibility loci are required. The most important population-specific genetic risk factor is the **Polynesian founder allele p.Glu386***, enriched in individuals of Polynesian (especially Samoan) ancestry.

Environmental "risk factors" (disease-precipitating triggers): Because baseline health is preserved, disease manifestation is triggered by specific environmental exposures: - **Live attenuated vaccines:** MMR, yellow fever 17D (YF-17D). - **Wild-type viruses:** SARS-CoV-2 (critical COVID-19 pneumonia), and other viruses in individual reports.

Protective factors: The principal "protective" measure is behavioral/medical — **avoidance of live attenuated vaccines**. No genetic protective modifier alleles have been established for this disease.

Gene–environment interaction: This disease is a paradigmatic **monogenic gene × environment interaction** — an inherited receptor defect that is clinically silent until an environmental viral trigger (vaccine or wild-type virus) unmasks it, producing life-threatening disease.

3. Phenotypes

The phenotype is dominated by **infectious/post-vaccination events** rather than constitutional features. Between triggers, patients are typically asymptomatic and grow normally.

Phenotype	Type	HPO suggestion	Onset	Severity	Frequency
Adverse reaction to live attenuated vaccine (MMR, YF)	Clinical event	HP:0002090 (Pneumonia), HP:0002383 (Encephalitis)	Childhood (post-vaccination)	Severe/life-threatening	Presenting feature in index cases
Viral encephalitis	Clinical sign	HP:0002383	Variable	Severe	Reported
Severe/critical viral pneumonia (incl. COVID-19)	Clinical sign	HP:0002090	Any age	Severe	Reported
Disseminated viral infection / viral hepatitis	Clinical sign	HP:0006562 (Viral hepatitis)	Variable	Severe	Reported
Susceptibility to viral infection	Lab/clinical	HP:0032251 (Abnormal susceptibility to viral infection)	Variable	Variable	Core
Otherwise normal immunity/health between triggers	—	—	—	—	Characteristic

Onset: Typically childhood, coincident with the routine live-vaccine schedule; can also present in adulthood upon wild-type viral challenge (e.g., adult critical COVID-19).

Severity/progression: Episodic and trigger-dependent. Individual episodes can be fulminant and fatal, but there is no constitutive, progressive organ degeneration between events.

Quality-of-life impact: Between episodes, QoL is generally normal. The dominant burden is the risk of catastrophic disease upon exposure and the lifelong requirement to avoid live vaccines and manage viral exposures. No disease-specific EQ-5D/SF-36 data are available (reflecting rarity).

"Vaccination against measles, mumps, and rubella (MMR) and yellow fever (YF) with live attenuated viruses can rarely cause life-threatening disease." — [PMID: 31270247](#)

4. Genetic / Molecular Information

Causal gene: *IFNAR1* (interferon alpha and beta receptor subunit 1), HGNC:5432, located at chromosome **21q22.11**. Gene OMIM 107450; disease phenotype OMIM #619304.

Pathogenic variants (spectrum):

Variant	Type	Population	Consequence	Reference
p.Glu386* (nonsense)	Truncating LOF	Polynesian founder (MAF >1% in Samoa)	Truncated protein absent from cell surface	PMID: 35442418
Large homozygous deletion	Structural / CNV	Individual case	Complete LOF	PMID: 35091979
CNV Δ3-4-5 (homozygous)	Structural / CNV	Brazilian family	Receptor dysfunction; fatal YF vaccine adverse event	PMID: 42097348

Variant classification: Reported disease-causing variants are **pathogenic** (nonsense, large deletions, CNVs) per ACMG/AMP, all producing loss of function.

Functional consequence: Loss of function — abolished expression or surface localization of *IFNAR1*, eliminating the type I IFN receptor complex. The founder allele encodes a **truncated protein absent from the cell surface**.

Allele frequency: The founder p.Glu386* allele is >1% MAF in Samoa but is extremely rare or absent in non-Polynesian populations. Other variants are private/ultra-rare.

Somatic vs germline: Germline. (Somatic *IFNAR1* down-regulation by viruses — e.g., SARS-CoV-2-induced *IFNAR1* ubiquitination [PMID: 34260266](#) — is a distinct immune-evasion phenomenon, not the inherited disease.)

Modifier genes / epigenetics / large chromosomal abnormalities: No established disease modifiers, epigenetic drivers, or aneuploidy associations specific to this monogenic disorder.

"All the patients are homozygous for the same nonsense IFNAR1 variant (p.Glu386). This allele encodes a truncated protein that is absent from the cell surface and is loss-of-function." — PMID: 35442418*

5. Environmental Information

Environmental/infectious triggers (the operative "environmental" dimension): - **Live attenuated vaccines:** measles-mumps-rubella (MMR), yellow fever 17D. - **Wild-type viruses:** SARS-CoV-2 (critical COVID-19 pneumonia); other viruses in individual reports.

Toxins/pollution/occupational exposures: Not implicated. **Lifestyle factors (smoking, diet, alcohol):** Not relevant to disease causation.

Infectious agents: The relevant "pathogens" are the **vaccine strains themselves** (attenuated measles, mumps, rubella, YF-17D) and specific wild-type viruses that normally depend on type I IFN for host containment.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic LOF variant in IFNAR1** (nonsense p.Glu386*, deletion, or CNV) → **leads to** absent or non-functional IFNAR1 protein at the cell surface.
2. Absent IFNAR1 → **prevents** assembly of the type I IFN receptor (IFNAR1 + IFNAR2) → **results in** no ternary receptor complex upon IFN- $\alpha/\beta/\omega$ binding.
3. No ternary complex → **fails to** juxtapose the receptor-associated kinases **JAK1 and TYK2** → they cannot cross-phosphorylate.
4. No JAK/TYK activation → **abolishes** phosphorylation of **STAT1/STAT2** → the **ISGF3 (STAT1–STAT2–IRF9)** complex does not form.

5. No ISGF3 → **eliminates** induction of **interferon-stimulated genes (ISGs)** → cells lose the type I IFN-driven antiviral state.
6. Loss of the antiviral state → **permits** uncontrolled replication of attenuated-vaccine viruses and select wild-type viruses in infected/target cells (e.g., fibroblasts highly vulnerable to SARS-CoV-2).
7. Uncontrolled viral replication → **causes** disseminated viral disease (encephalitis, pneumonitis, hepatitis, multiorgan involvement), which may be fatal.
8. *Branch (inferred)*: In some contexts, aberrant innate/inflammasome activation upon viral trigger contributes to immunopathology (e.g., inflammasome-driven innate-cell activation upon YFV-17D exposure in a CNV case [PMID: 42097348](#)).

Detail by category

Molecular pathway: The **type I IFN → JAK-STAT (ISGF3) → ISG** axis (KEGG hsa04630 Jak-STAT signaling; Reactome "Interferon alpha/beta signaling"). Type I IFNs bind a receptor formed by IFNAR1 and IFNAR2, forming a ternary complex that brings JAK1 and TYK2 into proximity; they cross-phosphorylate each other, the receptor chains, and STATs, which then activate ISGs [PMID: 38608537](#).

"Common to both are two distinct receptor chains (IFNAR1/IFNAR2 and IFNLR1/IL10R2), which form ternary complexes upon binding their respective ligands. This results in close proximity of the intracellularly associated kinases JAK1 and TYK2, which cross phosphorylate each other, the associated receptor chains, and signal transducer and activator of transcriptions, with the latter activating IFN-stimulated genes." — [PMID: 38608537](#)

Cellular processes: Loss of the cell-intrinsic antiviral state (impaired restriction of viral replication); GO:0060337 (type I interferon signaling pathway), GO:0051607 (defense response to virus).

Protein dysfunction: Loss of function of IFNAR1 — either the protein is truncated and fails to reach the plasma membrane (p.Glu386*) or is deleted entirely.

Immune system involvement: This is an **immunodeficiency** (impaired intrinsic antiviral immunity), not primarily an autoimmune or autoinflammatory disease — though excessive innate/inflammasome activation can accompany a viral trigger. Notably, the *opposite* pole of IFN biology (excessive IFN-I signaling) causes type I interferonopathies, underscoring the pathway's dose-sensitivity.

Species contrast (why the human phenotype is narrow): In mice, type I IFN is essential against a broad range of viruses; in humans there is far more redundancy under natural conditions [PMID: 33729549](#).

"A picture is emerging of greater redundancy of human type I IFNs for protective immunity to viruses in natural conditions than was initially anticipated." — [PMID: 33729549](#)

Molecular profiling / functional readout: IFNAR1-deficient fibroblasts show **no ISG induction** to IFN- $\alpha 2/-\beta/-\omega$ and are **highly vulnerable to SARS-CoV-2** ($P = 1.2 \times 10^{-11}$ vs benign controls) [PMID: 35708626](#).

"Fibroblasts deficient for IFNAR1, STAT2, or TYK2 are highly vulnerable to SARS-CoV-2." — [PMID: 35708626](#)

Cell types / GO / CL suggestions: Broadly acting because IFNAR1 is ubiquitously expressed — fibroblasts (CL:0000057), epithelial cells (CL:0000066), leukocytes/monocytes (CL:0000576), and other nucleated cells. GO terms: GO:0060337 (type I interferon signaling pathway), GO:0034340 (response to type I interferon), GO:0051607 (defense response to virus).

7. Anatomical Structures Affected

Because IFNAR1 is expressed on virtually all nucleated cells, the disease has **no fixed anatomical target**; rather, the organ affected is dictated by the tropism of the triggering virus.

- **Organ level:** Variable and virus-dependent — **lung** (viral/COVID-19 pneumonia; UBERON:0002048), **brain/CNS** (encephalitis; UBERON:0000955), **liver** (viral hepatitis; UBERON:0002107), with potential multiorgan dissemination.

- **Body systems:** Immune system (primary defect), with secondary respiratory, nervous, and hepatobiliary involvement during infectious episodes.
 - **Tissue/cell level:** Any infected cell type; fibroblasts and epithelial cells are demonstrably permissive in vitro. CL terms: CL:0000057 (fibroblast), CL:0000066 (epithelial cell), CL:0000576 (monocyte).
 - **Subcellular level:** IFNAR1 is a **plasma membrane** receptor (GO:0005886); signaling proceeds via cytoplasmic kinases/STATs to the **nucleus** (GO:0005634) for ISG transcription.
 - **Localization / lateralization:** Determined by viral tropism; typically bilateral/systemic in disseminated disease.
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8. Temporal Development

Onset: Congenital genetic defect, but **clinically silent until a triggering exposure**. Classic presentation is in **childhood** at the time of routine live-vaccine administration; adult presentation occurs with wild-type viral challenge (e.g., adult critical COVID-19).

Onset pattern: **Acute** — fulminant illness following vaccination or infection.

Progression / course: **Episodic**, trigger-dependent. There is no constitutive progressive degeneration; between episodes patients are typically well. Individual episodes can progress rapidly to severe/fatal disease.

Disease duration: The underlying genetic condition is **lifelong**; the risk persists indefinitely and requires lifelong avoidance of live vaccines.

Remission: Recovery from a viral episode is possible with supportive care and viral clearance; there is no "remission" of the underlying genetic defect.

Critical periods: The **peri-vaccination window** (routine childhood immunization schedule) is the key period of vulnerability and the main opportunity for prevention (withholding live vaccines).

9. Inheritance and Population

Inheritance pattern: Autosomal recessive. Recurrence risk for siblings of an affected proband is 25%.

Penetrance/expressivity: At the molecular level, biallelic LOF fully abolishes type I IFN responses; **clinical** penetrance is **trigger-dependent** (an individual may remain well if never exposed to a triggering live vaccine or virus). Expressivity is variable, driven by which virus is encountered.

Epidemiology: Ultra-rare globally. Prevalence has not been formally established; knowledge is based on case reports and small series.

Founder effect / affected populations: A strong **Polynesian founder effect** — the p.Glu386* allele has MAF >1% in Samoa and is also present in the Cook, Society, Marquesas, and Austral islands, and Fiji; it is extremely rare or absent elsewhere. Seven children from five unrelated western Polynesian kindreds were homozygous for this variant [PMID: 35442418](#).

Consanguinity: Contributes to homozygosity in some families (as typical for AR IEIs), and to the appearance of private homozygous deletions/CNVs.

Sex ratio: No sex predilection (autosomal).

"All the patients are homozygous for the same nonsense IFNAR1 variant (p.Glu386)... this IFNAR1 variant has a minor allele frequency >1% in Samoa." — [PMID: 35442418](#)*

10. Diagnostics

Diagnostic approach combines genetics + function:

1. **Genetic testing:** Whole-exome (WES) or whole-genome (WGS) sequencing, or targeted **inborn errors of immunity (IEI) gene panels**, to identify biallelic *IFNAR1* LOF variants. **Copy-number/CNV analysis is essential** because large deletions and structural CNVs (e.g., the homozygous deletion and CNV $\Delta 3-4-5$) are recurrent mechanisms [PMID: 35091979](#); [PMID: 42097348](#).
2. **Functional confirmation:** Demonstrate **absent cellular responses to type I IFNs** (IFN- $\alpha 2$, IFN- β , IFN- ω) using fibroblast or whole-blood ISG-induction assays. A sensitive whole-blood

assay (e.g., IP-10/CXCL10 readout) can detect both inborn errors and the autoantibody phenocopy [PMID: 39312669](#).

Differential diagnosis — critical: The key acquired phenocopy is **neutralizing autoantibodies against type I IFNs**, present in ~10–15% of critical COVID-19 pneumonia cases; these must be excluded [PMID: 42524028](#). Other differentials include IFNAR2, STAT1, STAT2, TYK2, IRF9 deficiencies and other IEIs of type I IFN immunity.

Screening: Not part of routine newborn screening; **cascade genetic screening** of relatives of affected individuals is recommended, particularly in high-prevalence Polynesian populations, prior to live vaccination.

"Human inborn errors of the type I IFN response pathway and auto-Abs neutralizing IFN- α , - β , and/or - ω can underlie severe viral illnesses. We report a simple assay for the detection of both types of condition." — [PMID: 39312669](#)

"Autoantibodies neutralizing type I interferon (AAN-I-IFN) have been found in at least 10-15% of critical COVID-19 pneumonia cases." — [PMID: 42524028](#)

11. Outcome / Prognosis

Mortality: Individual triggering episodes can be **fatal**. Documented fatalities include a child with concurrent critical COVID-19 pneumonia and MIS-C who died on day 56 [PMID: 35091979](#), and two deceased siblings following yellow fever vaccination in a Brazilian family with a homozygous *IFNAR1* CNV [PMID: 42097348](#).

Between-episode outlook: Generally good — patients are otherwise healthy with normal growth and development, and can survive to adulthood if triggers are avoided.

Complications: Encephalitis, viral pneumonia/ARDS, hepatitis, disseminated viral infection, and — in the COVID-19 setting — co-occurring MIS-C.

Prognostic factors: The nature of the trigger (fulminant vaccine-strain disease vs manageable wild-type infection), timeliness of recognition/supportive care, and avoidance of further live-vaccine exposure. With appropriate prevention (no live vaccines) and prompt management of infections, long-term prognosis can be favorable.

12. Treatment

There is no approved disease-specific pharmacotherapy. Critically, **exogenous type I IFN is ineffective** because the receptor is absent — this is a mechanistic dead-end for IFN-replacement strategies.

Modality	Role in IFNAR1 deficiency	NCIT suggestion
Supportive/intensive care	Mainstay during viral episodes (organ support, ICU)	NCIT:C15320 (Supportive Care)
Antiviral therapy	Directed against the triggering virus where available	NCIT:C258 (Antiviral Agent)
Hematopoietic stem cell transplantation (HSCT)	Rational curative option in severe cases (restores IFNAR1-competent immune cells)	NCIT:C15431 (Hematopoietic Stem Cell Transplantation)
Exogenous type I IFN	Not applicable — receptor absent	—

Pharmacogenomics / advanced therapeutics: No established gene therapy, RNA therapy, or targeted small-molecule therapy exists for this disease as of this report. HSCT is the only potentially curative intervention and is considered on a case-by-case basis in severe presentations.

Treatment strategy: Prevention-first (avoid live vaccines) + aggressive supportive/antiviral management of breakthrough viral illness + consideration of HSCT for severe/recurrent disease.

13. Prevention

Prevention is the cornerstone of management.

- **Primary prevention: Strict avoidance of live attenuated vaccines** (MMR, yellow fever, and other live vaccines) in affected individuals. Use inactivated/subunit alternatives where available and appropriate.

- **Secondary prevention: Cascade genetic screening** of at-risk relatives; and — importantly — **pre-vaccination screening** for type I IFN inborn errors and autoantibodies in relatives of individuals with adverse events following yellow fever vaccination [PMID: 42097348](#).
- **Population-targeted consideration:** Inherited IFNAR1 deficiency should be **considered in individuals of Polynesian ancestry with severe viral illnesses**, and consideration given before live vaccination in that population [PMID: 35442418](#).
- **Genetic counseling:** AR inheritance with 25% sibling recurrence risk; counsel families accordingly, including reproductive options.
- **Tertiary prevention:** Prompt recognition and supportive/antiviral care to limit complications during viral episodes.

"Our findings advocate for precision vaccinology by screening relatives of AEFI-YF cases for type I IFN EIs and auto-antibodies prior to live-attenuated vaccination." — [PMID: 42097348](#)

"Inherited IFNAR1 deficiency should be considered in individuals of Polynesian ancestry with severe viral illnesses." — [PMID: 35442418](#)

14. Other Species / Natural Disease

- **Orthologous gene:** Mouse *Ifnar1* (NCBI Gene ID 15975); human *IFNAR1* (NCBI Gene ID 3454). The type I IFN receptor architecture and JAK-STAT signaling are **evolutionarily conserved** across vertebrates, including teleost fish [PMID: 35906001](#).
- **Natural disease in other species:** No well-characterized naturally occurring **IFNAR1-deficiency disease** in companion animals or wildlife is established; the condition is defined in humans. Viral immune-evasion strategies that *degrade or block* IFNAR1 are, however, widespread across animal pathogens (e.g., African swine fever virus p22 promoting TAX1BP1-mediated IFNAR1 degradation [PMID: 40668839](#); lumpy skin disease virus LSDV122 disrupting IFNAR1/IFNAR2 assembly [PMID: 41525414](#)), underscoring the conserved centrality of this receptor to antiviral defense.
- **Evolutionary conservation of mechanism:** High — the requirement for type I IFN/JAK-STAT signaling in antiviral immunity is conserved, though with a **species difference in breadth**: mouse type I IFN is essential against a broad range of viruses, whereas human type I IFN shows greater redundancy [PMID: 33729549](#).

15. Model Organisms

The principal model is the **Ifnar1-knockout (IFNAR1 KO) mouse** (mammalian, germline knockout), one of the most widely used tools in viral pathogenesis and vaccine research.

Feature	IFNAR1 KO mouse	Human IFNAR1 deficiency
Type I IFN response	Abolished	Abolished
Viral susceptibility breadth	Broad — uniformly lethal to many viruses	Narrow — mainly live vaccines + select wild-type viruses
Utility	Pathogenesis + vaccine-efficacy studies	—

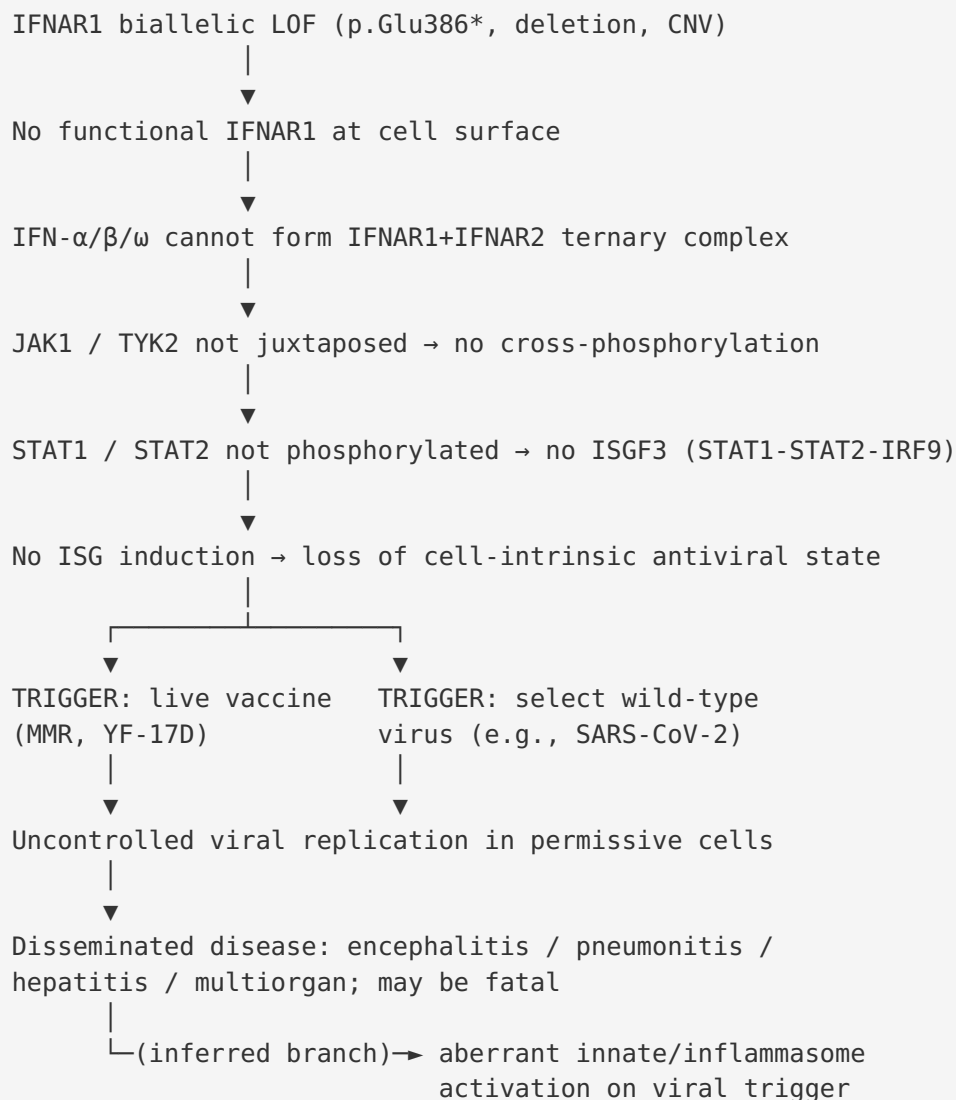
Phenotype recapitulation: The KO faithfully reproduces the **loss of type I IFN signaling** and its consequence (viral susceptibility). It is used as a **uniformly lethal infection model** across diverse viruses — Nipah [PMID: 42623407](#); [PMID: 42035922](#), Akabane [PMID: 40209629](#), Zika [PMID: 41754520](#), dengue [PMID: 42149728](#), Bourbon [PMID: 40910687](#), and other bunyaviruses — and for vaccine-efficacy testing.

Model limitation: The KO **overstates the human phenotype**. Because human type I IFN is more redundant than mouse type I IFN, KO mice are broadly and lethally susceptible to viruses that cause little or no everyday illness in IFNAR1-deficient humans [PMID: 33729549](#). Related multi-deficiency models (e.g., triple IFNAR/IFNGR/IFNLR "AGL" mice [PMID: 42463654](#)) extend susceptibility further and reveal type III IFN as a backup layer.

"Mouse type I IFNs are essential for protection against a broad range of viruses in experimental conditions." — [PMID: 33729549](#)

Resources: MGI (*Ifnar1*), IMPC/IMSR for knockout lines; patient-derived fibroblasts and iPSCs for in vitro functional assays.

Mechanistic Model / Interpretation



The unifying interpretation is that IFNAR1 deficiency **removes a single, non-redundant node** (the obligate IFNAR1 chain) of the type I IFN receptor, collapsing the entire IFN- $\alpha/\beta/\omega$ arm of antiviral immunity. In humans, this arm turns out to be **essential but narrow** — dispensable for containment of most everyday viruses (thanks to redundancy with type II/III IFN and other pathways), but indispensable for controlling attenuated-vaccine viruses and a limited set of wild-type viruses. This explains the paradox of an "otherwise healthy" patient who nonetheless suffers catastrophic vaccine or COVID-19 disease. The parallel with **IFNAR2 deficiency** (fatal encephalitis after MMR, yet no prior heightened respiratory-virus susceptibility [PMID: 26424569](#)) confirms the pattern applies to the whole IFNAR receptor.

"Despite the severe outcome of systemic live vaccine challenge, the proband had previously shown no evidence of heightened susceptibility to respiratory viral pathogens... supports an essential but narrow role for IFN- α/β in human antiviral immunity." — PMID: 26424569

Evidence Base

PMID	Study	Contribution	Source type
35442418	Bastard et al. 2022	Defines AR IFNAR1 deficiency; Polynesian founder p.Glu386* (MAF >1% Samoa); abolished IFN responses	Human clinical + functional
31270247	Hernandez et al. 2019	First description; MMR/YF live-vaccine adverse reactions in otherwise healthy patients	Human clinical
42097348	Azamor et al. 2026	Homozygous CNV $\Delta 3-4-5$; fatal YF vaccine AEs; advocates pre-vaccination screening	Human clinical + functional
35091979	Abolhassani et al. 2022	Homozygous LOF deletion; critical COVID-19 + MIS-C; fatal	Human clinical
35708626	Zhang et al. 2022	Recessive type I IFN IEIs in ~10.7% of pediatric COVID pneumonia; IFNAR1-deficient fibroblasts vulnerable to SARS-CoV-2	Human clinical + in vitro
38608537	de Weerd et al. 2024	Structure–function of type I/III IFN receptor signaling (ternary complex, JAK/TYK/STAT)	Review / structural
33729549	Meyts & Casanova 2021	Human vs mouse type I IFN redundancy; explains narrow human phenotype	Review
26424569	Duncan et al. 2015	IFNAR2 deficiency; "essential but narrow" role; otherwise-healthy phenotype	Human clinical
39312669	Gervais et al. 2024	Whole-blood assay detecting both inborn errors and autoantibody phenocopies	Methods
42524028	Kholaiq et al. 2026	Anti-type I IFN autoantibodies in 10–15% of critical COVID-19 (acquired phenocopy)	Human clinical

Consistency across independent kindreds and continents (Polynesia, Brazil, Iran, and pediatric COVID cohorts) reinforces the core disease definition. The mouse-model literature (Nipah, Akabane, Zika, dengue, Bourbon) both supports the mechanistic centrality of type I IFN and, by its broad lethality, **usefully contrasts** with the narrower human phenotype.

Limitations and Knowledge Gaps

- **Ultra-rarity:** Total reported cases number in the low dozens; there are no formal prevalence/incidence estimates, no QoL instruments, and no natural-history cohorts. Epidemiology outside Polynesia is essentially unknown.
 - **Variant spectrum incompletely mapped:** Beyond the founder nonsense allele and a few deletions/CNVs, the full mutational landscape (missense, splice, regulatory) and genotype–phenotype correlations are undefined.
 - **Trigger–outcome uncertainty:** It is not fully resolved which wild-type viruses cause severe disease in IFNAR1-deficient humans versus which are contained by redundancy. The role of inflammasome/innate hyperactivation as a driver of immunopathology is inferred, not proven.
 - **Therapeutics:** No disease-specific therapy or gene therapy exists; HSCT experience is anecdotal, with outcomes not systematically reported.
 - **Modifiers:** No genetic/epigenetic modifiers of severity have been identified.
 - **Model translation gap:** IFNAR1-KO mice overstate susceptibility, complicating direct translation of preclinical antiviral/vaccine findings.
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Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** for IFNAR1 (and broader type I IFN pathway) deficiency to capture prevalence, natural history, triggers, outcomes, and HSCT results.
2. **Population screening in Polynesia:** Systematic carrier screening for p.Glu386* and integration into pre-vaccination protocols; quantify carrier frequency and true regional prevalence.
3. **Genotype–phenotype cataloging:** Curate all *IFNAR1* variants (with CNV detection) in ClinVar/IEI databases and correlate variant class with clinical severity and triggering virus.

4. **Standardized functional diagnostics:** Broaden validation and deployment of whole-blood ISG-induction assays (IP-10/CXCL10) that simultaneously flag inborn errors and neutralizing autoantibodies, enabling rapid differential diagnosis in severe viral illness.
 5. **Prospective evaluation of HSCT** as curative therapy in severe cases; and preclinical exploration of *IFNAR1* gene-addition/editing in patient iPSC-derived immune cells.
 6. **Precision vaccinology guidelines:** Formalize recommendations to withhold/replace live vaccines and to screen relatives of individuals with severe adverse events following live-attenuated (especially yellow fever) vaccination.
 7. **Refine model systems:** Complement IFNAR1-KO mice with humanized or conditional models, and patient-derived organoids/iPSCs, to better recapitulate the narrow human antiviral phenotype for therapeutic testing.
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Report compiled from 9 confirmed findings across 5 investigation iterations, drawing on 38 reviewed papers. Evidence types are annotated as human clinical, model organism, in vitro, or review throughout.
